Restaurant Data Analysis Report
Levels 1 and 2

Cognifyz Internship Tasks | Exploratory Data Analysis with Python (pandas, matplotlib)

1. Introduction

This report covers the work done in two Jupyter notebooks, **level1_tasks.ipynb** and **level2_tasks.ipynb**. Both analyse the same restaurant dataset (Zomato-style data, Dataset .csv). Level 1 covers basic exploratory analysis: cuisines, cities, price ranges and online delivery. Level 2 covers deeper analysis: rating distribution, cuisine combinations, geographic patterns and restaurant chains.

2. Dataset Overview

Item	Value
Rows (restaurants)	9,551
Columns	21
Duplicate rows	0
Restaurant ID	All 9,551 values are unique
Missing values	Only Cuisines has missing values (9 rows)
Cities	141
Most common country code	1 (India), 8,652 restaurants (about 90.6%)
Aggregate rating	Mean 2.67, median 3.2, range 0 to 4.9
Votes	Mean 156.9, median 31, maximum 10,934

Key columns used: Restaurant Name, City, Locality, Longitude, Latitude, Cuisines, Price range, Has Online delivery, Aggregate rating and Votes.

3. Methodology

Tools: Python, pandas (v3.0.3) for data handling, matplotlib for charts, run in Jupyter.

Data preparation and EDA (Level 1)

1. Loaded the CSV with pd.read_csv and inspected it with head(), shape, info, describe() and columns.
2. Converted Restaurant ID to string and Country Code to category, because they are labels, not quantities. Averages of these columns are meaningless.
3. Checked data quality with isnull().sum() and duplicated().sum().

Analysis techniques used

Technique	Used for
str.split(",") + explode() + str.strip()	Counting individual cuisines when one cell holds several
value_counts() and idxmax()	Frequencies, top cities, top cuisines
Dividing counts by total rows	Percentages of restaurants
groupby() with mean() and agg()	Average ratings per city, delivery group, cuisine combination and chain
pd.cut()	Grouping ratings into ranges (0-1, 1-2, 2-3, 3-4, 4-5)
Filtering Aggregate rating > 0	Excluding unrated restaurants (a rating of 0 means not yet rated)
Minimum-count filters (10 restaurants per cuisine combination, 5 outlets per chain)	Avoiding averages based on very few restaurants
Bar charts and scatter plots	Visualising distributions, locations and chain statistics

4. Level 1: Summary of Completed Tasks and Findings

Task 1: Top Cuisines

Found the three most common cuisines after splitting multi-cuisine cells, and calculated the share of all restaurants serving each.

Cuisine	Restaurants	% of all restaurants
North Indian	3,960	41.46%
Chinese	2,735	28.64%
Fast Food	1,986	20.79%

These percentages do not add up to 100% because one restaurant can serve several cuisines.

Task 2: City Analysis

- **City with the most restaurants:** New Delhi (5,473 restaurants, about 57% of the dataset).
- **Average rating per city:** calculated for all 141 cities. Highest averages: Inner City (4.90), Quezon City (4.80), Makati City (4.65), Pasig City (4.63), Mandaluyong City (4.62). Lowest: Faridabad (1.87), Noida (2.04), Mc Millan and Montville (2.40 each), New Delhi (2.44).
- **City with the highest average rating:** Inner City (4.90).

Task 3: Price Range Distribution

Built a bar chart of restaurants per price range (1 to 4) and calculated percentages.

Price range	% of restaurants
1 (lowest)	46.53%
2	32.59%
3	14.74%
4 (highest)	6.14%

Nearly half of all restaurants are in the cheapest category, and the share falls steadily as price rises.

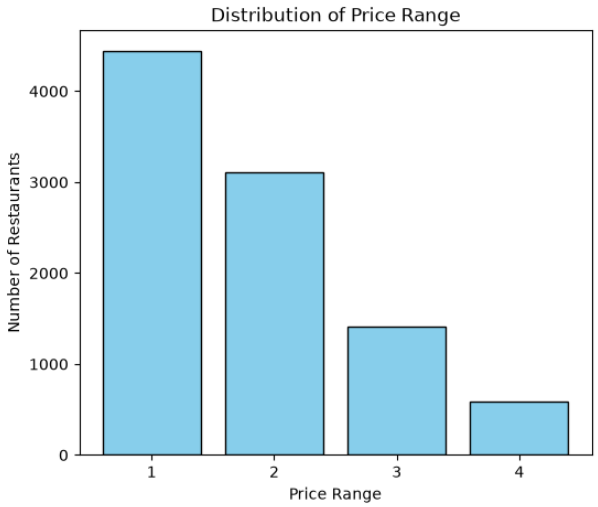


Figure 1: Distribution of price range (from level1_tasks.ipynb)

Task 4: Online Delivery

- **Share offering online delivery:** about 25.7% (2,451 of 9,551). This is derived from the notebook's own describe() output, which shows 7,100 "No" values. Note that the figure printed in the notebook (12.12%) is the table booking share, not delivery.

Online delivery	Average rating
No	2.47
Yes	3.25

Restaurants with online delivery have a clearly higher average rating (a gap of 0.78).

5. Level 2: Summary of Completed Tasks and Findings

Task 1: Restaurant Ratings

- **Distribution:** across all restaurants the mean rating is 2.67 and the median 3.2. The mean is pulled down by restaurants with a rating of 0 (unrated).
- **Most common rating range (rated restaurants only): 3-4**, with roughly 4,400 restaurants. Next are 2-3 (about 1,900) and 4-5 (about 1,100). The 0-1 and 1-2 ranges are almost empty. These counts are read from the bar chart, so they are approximate.
- **Average votes per restaurant:** 156.91. This is far above the median of 31, so a few very popular restaurants inflate the average and most restaurants receive far fewer votes than the mean suggests.

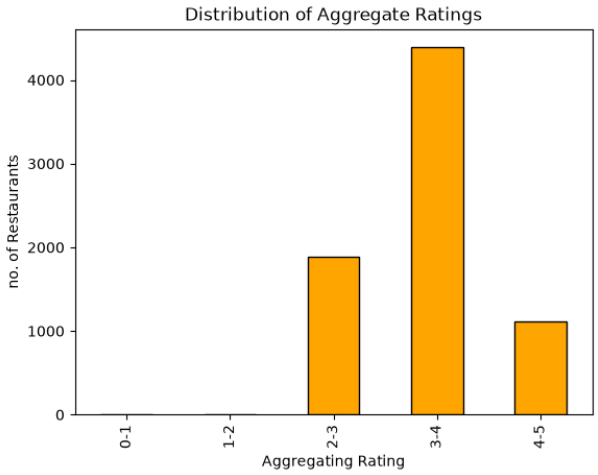


Figure 2: Distribution of aggregate ratings, rated restaurants only (from level2_tasks.ipynb)

Task 2: Cuisine Combinations

Most common combinations (exact text):

Combination	Restaurants
North Indian	936
North Indian, Chinese	511
Chinese	354
Fast Food	354
North Indian, Mughlai	334

Do some combinations rate higher? Yes. Using only rated restaurants and combinations with at least 10 restaurants (91 combinations qualified):

- Highest average ratings: Modern Indian (4.35), Indian (4.25), Seafood (4.11), Thai (4.10), American, Burger (4.08).
- Lowest average ratings: Biryani, North Indian (2.86), Raw Meats, Fast Food (2.99), American, Fast Food, Salad, Healthy Food (3.00), North Indian, South Indian, Chinese, Fast Food (3.00), North Indian, South Indian (3.02).

Specialised or premium-style cuisines tend to rate higher, while fast-food-heavy and generic multi-cuisine combinations rate lower. The most common combinations are not the highest rated.

Task 3: Geographic Analysis

Plotted all restaurants as a longitude/latitude scatter plot, then a second plot with invalid (0, 0) coordinates removed and semi-transparent points so dense areas look darker.

Pattern: the data spans the globe (longitude from about -158 to 175, latitude from about -41 to 56), but the middle 50% of restaurants lie in a very narrow band (longitude about 77.1 to 77.3, latitude about 28.5 to 28.6). That is the Delhi NCR region. The plot shows one very dense cluster there, with other cities appearing as small, scattered groups.

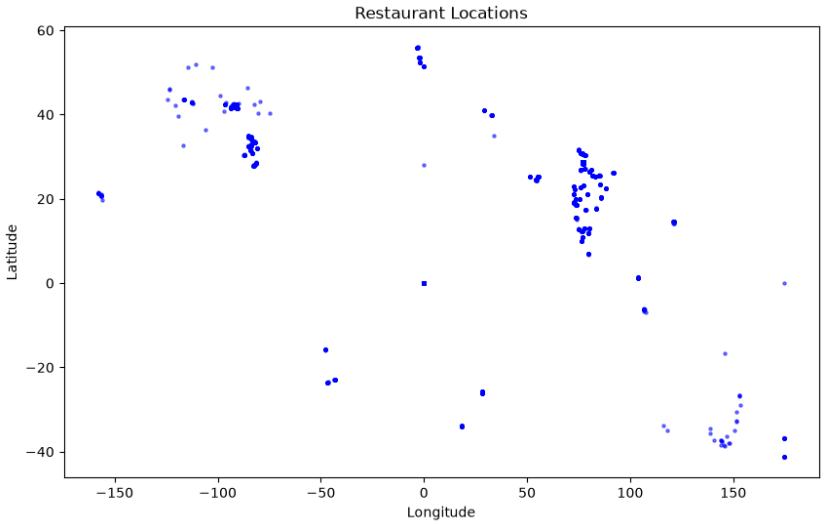


Figure 3: Restaurant locations by longitude and latitude (from level2_tasks.ipynb)

Task 4: Restaurant Chains

Are there chains? Yes. 734 restaurant names appear more than once. Largest chains by outlets:

Chain	Outlets
Cafe Coffee Day	83
Domino's Pizza	79
Subway	63
Green Chick Chop	51
McDonald's	48
Keventers	34
Pizza Hut	30
Giani	29
Baskin Robbins	28
Barbeque Nation	26

- Most popular by average votes: Truffles (about 4,800), Joey's Pizza, AB's - Absolute Barbecues, Big Chill and Big Yellow Door.
- Best rated chains with at least 5 outlets: Chili's (about 4.6), Farzi Cafe, Barbeque Nation, Punjab Grill and Mocha (all above about 4.2).
- **Insight:** being large does not mean being best. The biggest chains by outlet count (Cafe Coffee Day, Domino's, Subway) appear in neither top-10 list. Barbeque Nation is the one chain that is both large and highly rated.

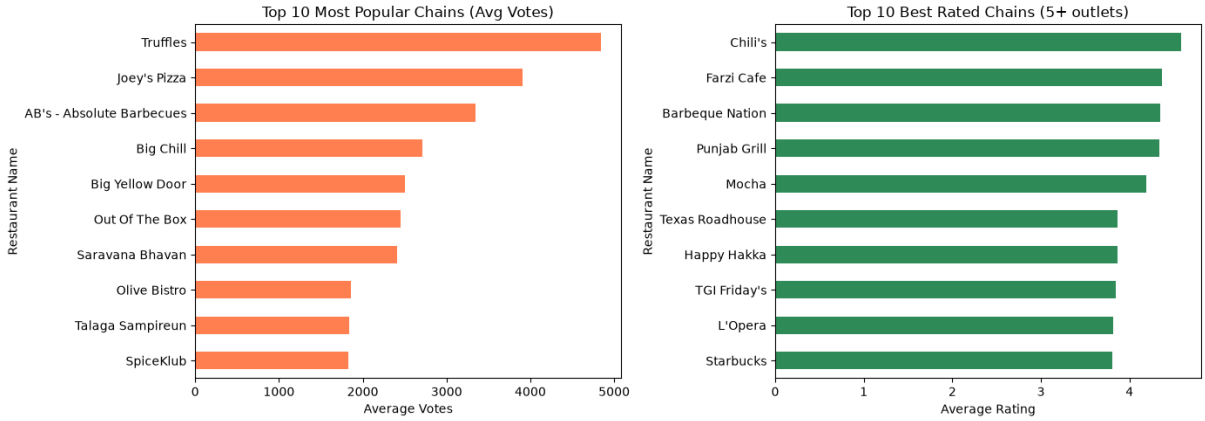


Figure 4: Most popular chains (average votes) and best rated chains with 5+ outlets (from level2_tasks.ipynb)

6. Learning Outcomes

- Performed a full EDA workflow in pandas: loading, inspecting structure, checking types, missing values and duplicates.
- Learned that numeric-looking columns (IDs, country codes) are really categories and should be converted so statistics stay meaningful.
- Learned to handle multi-valued text columns with split, explode and strip before counting.
- Became comfortable with groupby, agg, value_counts, idxmax and pd.cut to summarise and compare groups.
- Understood how a placeholder value (a rating of 0) can distort averages, and why small groups need a minimum-size filter.
- Learned that mean and median can differ widely in skewed data (Votes: mean 156.9, median 31).
- Practised choosing the right chart: bar charts for categories, scatter plots for coordinates, side-by-side charts for comparing two measures.
- Practised geographic visualisation with latitude and longitude, and cleaning invalid coordinates.
- Learned to separate popularity (votes) from quality (ratings) and to interpret both together.
- Learned to check results against the code and the data, as in the delivery-column mix-up, and to state limitations honestly.

7. Conclusion

The dataset is dominated by India, and by Delhi in particular. North Indian, Chinese and Fast Food are the most common cuisines, and almost half of the restaurants sit in the cheapest price band. Online delivery goes with higher ratings (3.25 versus 2.47). Most rated restaurants score between 3 and 4. Specialised cuisines such as Modern Indian, Seafood and Thai rate highest. The biggest chains are not the best rated or the most popular.